

3.5 Rational Numbers in Real-World Context

Lesson Introduction

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 Benchmark	MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite directions using rational numbers. Compare them on a number line and explain the meaning of zero within its context.		 Learning Targets	- I can determine positive and negative values in relation to real-world contexts. - I can plot negative rational values given a real-world context.	
 Benchmark Coherence	Building On		Working Toward		
	N/A		N/A		
 Essential Questions	- How do I interpret positive and negative rational numbers in real-world contexts? - How can a number line be used to plot positive and negative rational numbers in real-world contexts? - What is the meaning of zero in various real-world contexts?				
 Lesson Narrative	In this lesson, students will interpret positive and negative values in a real-world context, including plotting the values on a number line. Students interpret and plot values related to sea level, finance, weather, and football during the lesson.				
 Component Summary	3.5.1 Warm-Up (~5 min.)	Reflect on biography of a NASA mathematician (<i>SE p. 98</i>)	3.5.4 Collaboration (5-10 min.)	Collaboratively practice determining and plotting values related to temperature (<i>SE p. 101</i>)	
	3.5.2 Collaboration (5 min.)	Exploration of elevation with small groups (<i>SE p. 99</i>)	3.5.5 Try It (10 min.)	Practice plotting and interpreting rational numbers in real-world contexts (<i>SE p. 102</i>)	
	3.5.3 Guided Instruction (10 min.)	Students learn how to relate positive and negative values to the real-world contexts (<i>SE p. 100</i>)	3.5.6 Your Turn! (5 min.)	Additional practice interpreting rational numbers in real-world contexts (<i>SE p. 104</i>)	
	3.5.7 Wrap-Up (~5 min.)	Demonstrate understanding of interpreting and plotting real-world values (<i>SE p. 104</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Elevation - Deposit - Withdrawal - Expense <u>Additional Terms:</u> N/A		(Optional) Dry erase boards		3.5.1 Warm-Up contains a video to accompany the question prompt. - Prior to question 3 in 3.5.3 Guided Instruction, share pictures related to the Samuel L. Gravely Jr.'s important contributions to the Navy.

 Teacher Tips	Common Misconceptions - Students may incorrectly say both “negative” and “below” or “positive” and “above” when describing the elevation relative to sea level. - Students may not be able to compare Celsius temperatures to Fahrenheit temperatures or understand that 0°C, not 32°F, is the freezing point. - Students may get stuck with zero representing no difference from the monthly average rainfall in 3.5.5 Try It question 3.	Things to Consider - Facilitate a conversation about the use of the negative symbol in conjunction with words that represent negative values (i.e., below, loss, withdraw) to help students learn how to describe the meaning of values in context correctly. - Explain the Celsius temperature scale to students and show how to represent temperatures on a vertical number line. - Remind students to include the units (i.e., yards) in their answers.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share In 3.5.2 and 3.5.4 Collaborations, students initially complete the questions and then discuss their work with a partner. This discussion builds student confidence and understanding.	MA.K12.MTR.7.1: Real-World Contexts Throughout this lesson, students use real-world scenarios to interpret and plot positive and negative numbers. Students interpret and plot values related to sea level, finance, weather, and football.
	 Differentiate Instruction	Leverage the real-world contexts of this lesson for extensions and project-based learning ideas for your students as homework or as projects to demonstrate their understanding at this midpoint in the unit. 3.5.1 Warm-Up, Katherine Johnson: Students can show and demonstrate at least one (simplified) version of one of the ways Katherine Johnson would have used positive and negative rational values in context at NASA. 3.5.2 Collaboration or 3.5.3 Guided Instruction: Students can use coordinate planes or number lines to map and log the elevation of their favorite landmarks from their home country or home state. 3.5.4 Collaboration: Students can log and display the temperature of different countries during different seasons of the year.
Lesson Notes:		

3.5.1 Warm-Up: Mathematician Katherine Johnson

Component Overview: Students learn about and then reflect on the career of Katherine Johnson, a NASA mathematician.



3.5.1 Warm-Up: Mathematician Katherine Johnson

Born in 1918, Katherine Johnson was always fascinated by numbers. Graduating from high school at 14 and from



college at 18, she went on to become one of the first African American mathematicians with the job title of “computer” at NASA, where she worked for three decades. As a computer, she personally checked the calculations made by electronic computers. Katherine calculated the launch window for America’s first human space flight. Her calculations were critical to the success of the Apollo Moon landing program and the start of NASA’s Space Shuttle program. In 2015, she received the Presidential Medal of Freedom from President Obama. In its official announcement, the White House cited Katherine’s influence on every major space program from Mercury to the Shuttle Program.

1. What stands out to you the most about Katherine Johnson?

Answers will vary.



Do you know a mathematician, statistician, actuary, or engineer? Consider inviting them to be a guest speaker in your classroom. Additionally, consider having students return to this lesson later in the year for project-based learning around this career profile.

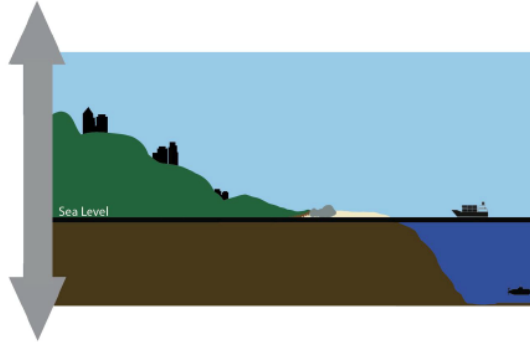
3.5.2 Collaboration: Exploring Elevation

Component Overview: Students work collaboratively as they explore how to determine and plot elevations above and below sea level. Use the image to show students how sea level represents zero and the distance above and below sea level can be represented by positive and negative numbers on a vertical number line.



3.5.2 Collaboration: Exploring Elevation

The picture shows different locations relative to sea level.



- Describe how the picture can be used to explain the word *elevation*.
Answers will vary. The grey arrow on the side shows the elevation of different things in the picture, since elevation is how high or low something is.
- Ask two classmates for their description of the word elevation. Write their description in the table. *You are the only person who should write in the first column of the table.* Have your classmate initial next to your summary of their description stating that your summary is correct.

My Summary of _____'s Description of Elevation	Initials
<i>Answers will vary.</i>	
My Summary of _____'s Description of Elevation	Initials

- Using the picture, name one thing that is below sea level.
Answers will vary. Examples: Dirt/Mud, water, submarine, etc.
- Name one thing that is above sea level.
Answers will vary. Examples: Grass, sky, ship, etc.



For some students, this may be the first time they are hearing or discussing the word elevation. Listen as students collaborate to determine who has prior knowledge of this concept or who develops understanding through their collaboration with peers in this component. The term will be formally defined in the next component, so allow students to focus on sharing what they already know about elevation without overemphasizing that they arrive at a "correct" definition in this component.



- What value do you think sea level represents on the number line?*
- What type of values are above sea level on the number line?*
- What type of values are below sea level on the number line?*

3.5.3 Guided Instruction: Positive and Negative Values in the Real World

Component Overview: Students learn to relate positive and negative rational numbers to the contexts of elevation and finance. Prior to question 2, share pictures related to the Samuel L. Gravely Jr.'s important contributions to the Navy.



3.5.3 Guided Instruction: Positive and Negative Values in the Real World

Elevation is the vertical distance a location or object is from a given level, such as sea level.

Elevation is represented using positive and negative values.

1. If sea level is equivalent to an elevation of 0 units, then what type of value is an elevation that is below sea level?

A negative value.

2. Write each situation as an integer. Plot it on the vertical number line.

- a. A hiker is 37 feet above sea level.

37

- b. A diver is 19 feet below sea level.

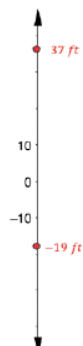
-19

- c. How are these two integers related?

Answers will vary. Both values represent position relative to sea level.

- d. What does 0 represent in this situation?

Sea level.



3. Samuel L. Gravely, Jr. pioneered the way with a plethora of firsts for African Americans in the military. Gravely was the first African American Navy Vice Admiral, first African American to command a Navy warship, first African American to command a warship during combat, first African American to command a Navy fleet, and first African American to obtain Flag Rank in the military. If the Navy fleet included a submarine that was submerged 780 feet below sea level, which students' interpretation of the situation is correct? Circle the name of the student with the correct answer. For any statement that has an error, write a corrected statement.



Consider having students read the vocabulary words aloud to provide an auditory entry point. A visual entry point is provided through the plotting of values on number lines.

3.5.3 Guided Instruction: Positive and Negative Values in the Real World (continued)

Tianna	Keagan
The depth of the submarine is -780 feet below sea level. <i>Answers will vary.</i> <i>The negative in front of 780 indicates that the submarine is below sea level, so it is not correct to say -780 below sea level. It should either be "-780 feet" or "780 feet below sea level," but not both.</i>	780 feet below sea level can be represented by the integer -780.

A bank **deposit** is the amount of money a person gives the bank for safekeeping in their account.

A bank **withdrawal** is the amount of money removed from a bank account.

An **expense** is a cost or the amount of money spent.

4. The following people have money in their bank accounts. Determine the positive and negative values that represent the effect each situation has on their account balance.

Real-World Context	Value
Paul paid \$190.90 in taxes.	$-\$190.90$
Marie made a deposit of \$50.	$\$50$
Tyler made a withdrawal of \$200.	$-\$200$
Hannah gets paid an income of \$500 per week that she deposits in her account.	$\$500$
Kaden pays \$758.91 in monthly expenses.	$-\$758.91$



This is the first unit where students are working with signed values. Use a think aloud to model how to answer question 3 since the students will not necessarily have the prior knowledge to recognize what is wrong with Tianna's response. They will have more practice with this concept over the next few lessons.



Encourage students to create a reference card with vocabulary words and examples. Students can use the reference cards to complete the subsequent question independently.

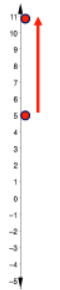
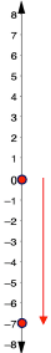
3.5.4 Collaboration: Above or Below Zero

Component Overview: Students collaborate to interpret the meaning of temperature changes using a number line. Students must model that change on the number line in addition to answering a question about the line to check for understanding regarding plotting values.



3.5.4 Collaboration: Above or Below Zero

1. Represent each change in temperature on the number line. Answer the question for each situation.

At noon, the temperature was 5 degrees Celsius. By late afternoon, the temperature had risen by 6 degrees Celsius.		What was the temperature late in the afternoon? <i>11 degrees Celsius</i>
The temperature was 8 degrees Celsius at midnight. By dawn, the temperature had dropped by 12 degrees Celsius.		What was the temperature at dawn? <i>-4 degrees Celsius</i>
Water freezes at 0 degrees Celsius, but the freezing temperature can be lowered by adding salt to the water. A student discovered that adding half a cup of salt to a gallon of water lowers its freezing temperature by 7 degrees Celsius.		What is the freezing temperature of the gallon of salt water? <i>-7 degrees Celsius</i>



If necessary, explain the Celsius temperature scale to students and show how to represent temperatures on a vertical number line.



Students initially complete the questions and then discuss their work with a partner. This discussion builds student confidence and understanding.

3.5.4 Collaboration: Above or Below Zero (continued)

Numbers below 0 are useful in contexts other than temperature.

2. With your partner, discuss and list other examples where numbers below zero can be used to represent a situation.

Answers will vary. Examples: elevation, money in a bank account.



Expect answers from students such as elevation (“below sea level”) and bank accounts.

3.5.5 Try It: Representing the Real World With Positive and Negative Values

Component Overview: Students practice interpreting and plotting real-world values related to finance, football, and rainfall with a partner. Circulate while students work to support struggling students, monitor student progress, and select which students you will have share during the debrief.



3.5.5 Try It: Representing the Real World with Positive and Negative Values

1. In a bank account, every time money is added or taken out of the account, a new balance is calculated. What does each of the following mean?
 - a. an account balance of $-\$23.56$
Answers will vary.
The account is overdrawn by \$23.56.
 - b. an account balance of $\$143.21$
Answers will vary.
There is \$143.21 available in the account.
 - c. an account balance of $\$0$
Answers will vary.
There is no money in the account.
2. The change in position of the ball during each play of a football game is measured in yards.
 - a. What integer best represents a gain of 5 yards?
+5
 - b. What integer best represents a loss of 15 yards?
-15
 - c. What does the number 0 represent in this context?
Answers will vary.
No loss or gain of yards.
 - d. Plot all three values on the number line.



If students struggle determining if the numeric value is positive or negative, prompt them to identify the key terms. Ask students what clues the terms provide regarding the sign of the number.

3.5.5 Try It: Representing the Real World With Positive and Negative Values (continued)

3. The average monthly rainfall for a city was calculated based on records kept over the past 50 years. The chart shows the difference between the average monthly rainfall and the rainfall amount this year for these months.

Month	Difference From Monthly Average (in inches)
January	$-2\frac{3}{4}$
February	$1\frac{5}{8}$
March	0
April	$-\frac{3}{8}$
May	$4\frac{1}{4}$



Which month had the greatest change from the average monthly rainfall, and how do you know? What does zero represent in the rainfall chart?

Explain what the value in the chart means for the following months in the context of this problem. Then, plot and label each month's value on the number line.

Month	Interpretation	
February	<i>Answers will vary. In February, it rained $1\frac{5}{8}$ inches more than the average rainfall typical for February.</i>	
March	<i>Answers will vary. In March, it rained exactly the typical average amount for March.</i>	
April	<i>Answers will vary. In April, it rained $\frac{3}{8}$ less than the average rainfall typical for April.</i>	

3.5.6 Your Turn!

Component Overview: Students independently practice interpreting rational numbers in real-world contexts. Be sure to remind students to include units in their responses.



3.5.6 Your Turn!

1. Write a value that represents each description.

- | | |
|-----------------------------------|----------------|
| a. an expense of \$40.23 | <u>-40.23</u> |
| b. a deposit of \$225.89 | <u>225.89</u> |
| c. 16,000 feet above sea level | <u>16,000</u> |
| d. a temperature increase of 14°F | <u>14</u> |
| e. a withdrawal of \$271.49 | <u>-271.49</u> |
| f. a gain of 17 yards | <u>17</u> |

2. Read each statement and select the correct description of each situation.

a. A diver is 60 feet below the surface of the ocean.

- ☐ The depth of the diver is 60 feet below the ocean's surface.
- ☐ The diver is -60 feet below the ocean's surface.

b. The elevation of the bottom of a boat with respect to sea level is given as -5 feet.

- ☐ The bottom of the boat is 5 feet above sea level.
- ☐ The bottom of the boat is 5 feet below sea level.



Do your students learn best by co-creating their learning contexts? Consider offering students the opportunity to create their own sentence frames with two options to choose from, then trade with a partner.

3.5.7 Wrap-Up: Ticket Out the Door

Component Overview: In this formative assessment, students demonstrate their understanding of interpreting and plotting rational numbers in real-world contexts.



3.5.7 Wrap-Up: Ticket Out the Door

Keisha observes that one location is 55.2 feet above sea level and another is 46.8 feet below sea level.

1. What rational number represents a location of 55.2 feet above sea level?

55.2 or $55\frac{2}{10}$ or $55\frac{1}{5}$

2. What rational number represents a location of 46.8 feet below sea level?

-46.8 or $-46\frac{8}{10}$ or $-46\frac{4}{5}$

3. Plot and label each rational number on the number line.



4. What does zero represent in this situation?

Sea level



Encourage students to share their reasoning for what sea level represents by depicting their interpretations of the situation. Prompt students to create what each location could actually represent, as context could contribute to better conceptual understanding.